

MODEL QUESTION PAPER SET I**Analytic Instruments****Part A**

Answer all questions in one word or one sentence.

- 1 Explain Beer Lamberts Law?
2. Define sensitivity?
3. Name the different types of Spectrophotometer?
4. Name the types of detectors used in IR spectroscopy?
5. Give any two applications of Flame Emission Spectroscopy?
6. Define Chromatography?
7. Write one advantage of gas chromatography?
8. Write the equation for pH calculation based on hydrogen ion concentration?
9. A is a small, portable machine that's used to measure how much glucose (a type of sugar) is present in the blood

1* 9=9 marks

PART B

II. Answer any eight questions from the following, Each question carries 3marks.

- 1.List any 6 analytical instruments used in clinical environment?
- 2.Explain any three the applications of colorimeter?
- 3.Describe electromagnetic Spectrum?
- 4.Describe the principle of absorption spectroscopy?
- 5.Explain the constructional details of flame photometer?
- 6.List the classifications of Chromatography?
- 7 Explain the applications of spectrophotometer in biochemical analysis?
8. Describe electrophoresis apparatus with a block diagram?
9. Describe the construction details of pH meter?

10. Explain the principle of glucometer?

3* 8=24 marks

PART C

Answer ALL questions. Each question carries 7 marks

III Explain the elements of analytical instrument?

OR

IV Explain principle & working of colorimeter ?

V Draw and explain the schematic diagram of double beam spectrophotometer?

OR

VI Describe different radiation source used in absorption spectrophotometry?

VII With a block diagram explain gas chromatography?

OR

VIII With a neat schematic diagram discuss the separation principle of HPLC?

IX Explain the principle and construction of PO₂ electrode?

OR

X With a suitable block diagram explain Hb meter?

XI Illustrate blood gas analyzer with suitable block diagram ?

OR

XII Explain the working of autoanalyzer with block diagram?

XIII Explain working of IR spectroscopy

OR

XIV Describe principle of flame photometry.

7* 6=42 marks

MODEL QUESTION PAPER SET II**Analytic Instruments****Part A**

Answer all questions in one word or one sentence.

1. List the law relating to absorption of radiation?
2. Define resolution?
3. What is spectroscopy?
4. Name the types of detectors used in IR spectroscopy?
5. Give any two applications of Spectrophotometer?
6. List the classifications of Chromatography?
7. Write one application of HPLC?
8. List the electrodes used in pH meter?
9. A is an apparatus for counting and sizing particles suspended in electrolytes.

1* 9=9 marks

PART B

II. Answer any eight questions from the following, Each question carries 3marks.

1. Differentiate between accuracy, precision and sensitivity?
2. Explain calibration procedure of colorimeter?
3. List the elements of analytic Instruments?
4. Describe the principle of absorption spectroscopy?
5. Explain the constructional details of UV spectrophotometer?
6. List the classifications of Chromatography?
7. Explain any three applications of flame photometer?
8. Describe densitometer with a block diagram?
9. Describe the construction details of pH meter?

10. Explain the block diagram of glucometer?

3* 8=24 marks

PART C

Answer ALL questions. Each question carries 7 marks

III Explain PC based analytical instrument?

OR

IV Explain principle & working of colorimeter ?

V Draw and explain the schematic diagram of single beam spectrophotometer?

OR

VI Describe different radiation source used in absorption spectrophotometry?

VII With a block diagram explain liquid chromatography?

OR

VIII With a neat schematic diagram discuss the separation principle of HPLC?

IX Explain the principle and construction of PCO₂ electrode?

OR

X With a suitable block diagram explain blood cell counter?

XI Illustrate pH meter with suitable block diagram ?

OR

XII Explain the working of autoanalyzer with block diagram?

XIII Explain working of visible spectroscopy

OR

XIV Describe construction details of flame photometry.

7* 6=42 marks